

```
create database It_monitoring
```

```
use It_monitoring
```

```
-- Table 1: Servers
```

```
--select * from Servers
```

```
CREATE TABLE Servers (
```

```
server_id INT PRIMARY KEY,
```

```
server_name VARCHAR(50),
```

```
region VARCHAR(50)
```

```
);
```

```
--drop table Servers
```

```
--SELECT * FROM dbo.Servers WHERE server_id = 55;
```

```
--SELECT * FROM Servers
```

```
INSERT INTO Servers VALUES
```

```
(1, 'web-server-01', 'us-east'),
```

```
(2, 'db-server-01', 'us-east'),
```

```
(3, 'api-server-01', 'eu-west'),
```

```
(4, 'cache-server-01', 'us-west');
```

```
-- Table 2: Alerts
```

```
CREATE TABLE Alerts (
```

```
alert_id INT PRIMARY KEY,
```

```
server_id INT,  
alert_type VARCHAR(50),  
severity VARCHAR(20)  
Foreign Key (server_id) references Servers(server_id),  
);
```

```
SELECT * FROM Alerts
```

```
--drop table Alerts
```

```
INSERT INTO Alerts VALUES  
(101,1, 'CPU Spike', 'High'),  
(102,2, 'Disk Failure', 'Critical'),  
(103, 3, 'Memory Leak', 'Medium'),  
(104,4, 'Network Latency', 'Low'); -- Invalid server_id (edge case)
```

```
-- Table 3: AI Models
```

```
CREATE TABLE AI_Models (  
model_id INT PRIMARY KEY,  
model_name VARCHAR(50),  
use_case VARCHAR(50)  
);
```

```
INSERT INTO AI_Models VALUES  
(201, 'AnomalyDetector-v2', 'Alert Prediction'),  
(202, 'ResourceForecaster', 'Capacity Planning'),
```

```
(203, 'LogParser-NLP', 'Log Analysis');
```

```
-- Table 4: ModelDeployments
```

```
CREATE TABLE ModelDeployments (  
  deployment_id INT PRIMARY KEY,  
  server_id INT,  
  model_id INT,  
  deployed_on DATE  
  Foreign Key (server_id) references Servers(server_id),  
  Foreign Key (model_id) references AI_Models(model_id)  
);
```

```
--drop table ModelDeployments
```

```
INSERT INTO ModelDeployments VALUES
```

```
(301, 1, 201, '2025-06-01'),
```

```
(302, 2, 201, '2025-06-03'),
```

```
(303, 2, 202, '2025-06-10'),
```

```
(304, 3, 203, '2025-06-12');
```

```
--select alert_id, server_id
```

```
--From Alerts A, Servers S
```

```
--Where A.alert_id = S.server_id
```

```
--task 1 inner join
```

```
select alert_id, server_id
```

```
From Alerts A,Servers S ,
```

```
where S.server_id = A.alert_id
```

```
--Task 1: INNER JOIN
```

```
SELECT Alerts.server_id,alert_id
```

```
FROM Alerts
```

```
INNER JOIN Servers ON Alerts.server_id = Servers.server_id;
```

```
SELECT Alerts.server_id,alert_id,alert_type,severity
```

```
FROM Alerts
```

```
INNER JOIN Servers ON Alerts.server_id = Servers.server_id;
```

```
--Task 2:LEFT JOIN
```

```
SELECT s.server_id, s.server_name, a.alert_id, a.alert_type, a.severity
```

```
FROM Servers s LEFT JOIN Alerts a
```

```
ON s.server_id = a.server_id
```

```
--Task 3:RIGHT JOIN
```

```
SELECT s.server_id, s.server_name, a.alert_id, a.alert_type, a.severity
```

```
FROM Servers s RIGHT JOIN Alerts a
```

```
ON s.server_id = a.server_id
```

```
--Task 4: FULL OUTER JOIN
```

```
SELECT Servers.server_id,server_name,region,alert_id,alert_type,severity,Alerts.server_id
```

```
FROM Servers
```

```
Full Join Alerts ON Alerts.alert_id = Servers.server_id;
```

```
--Task 5: CROSS JOIN
```

```
SELECT S.server_id,S.server_name,M.model_id,M.model_name,M.use_case  
FROM Servers S CROSS JOIN AI_Models M;
```